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A systematic review protocol examining endothelial function and carotid artery disease in patients with inflammatory bowel disease

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Keywords:	Inflammatory bowel disease < GASTROENTEROLOGY, Vascular medicine < INTERNAL MEDICINE, CARDIOLOGY

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A systematic review protocol examining endothelial function and carotid artery disease in patients with inflammatory bowel disease

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Abstract

Introduction: Patients with inflammatory bowel disease (IBD) might be at an increased risk for the development of cardiovascular disease (CVD). The present protocol is developed to review and analyze published data to determine if patients with IBD have an increased CVD burden.

Methods and analysis: We will conduct a systematic review of all observational studies that examine endothelial function, arterial stiffness, and carotid intima-media thickness in IBD patients. Study selection will follow the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, and study quality will be assessed using the Newcastle-Ottawa Scale (NOS). If sufficient data are available, a meta-analysis will be conducted. The overall effect sizes will be estimated using both fixed-effects models and random-effects models. Statistical heterogeneity will be calculated using Higgin's (I^2) tests. Subgroup analyses, conditional number of studies retrieved and their sample size, will be stratified according to participant disease category or gender or disease activity.

Ethics and dissemination: Formal ethics approval is not required as individual data will not be collected. The results will be disseminated through peer-reviewed publications, conference presentations and scientific news releases.

Trial registration number: International Prospective Register for Systematic Reviews (PROSPERO) number CRD42021274093.

Strengths and limitations of this study

- Will provide an up-to-date and comprehensive review on the data on endothelial dysfunction and CVDs in patients with IBD.
- The data reporting will adhere to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.
- The study may be impacted by significant heterogeneity in the included studies due to differences in diagnostic tests, disease activities, and the presence of traditional cardiovascular risk factors.

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Introduction

Rationale: Cardiovascular disease (CVD) remains the leading cause of morbidity and mortality despite aggressive treatment of traditional risk factors (1). Systemic inflammation and endothelial cell dysfunction are considered among the key factors that are critically involved in the development and progression of CVD (2, 3). Indeed, patients with chronic inflammatory diseases such as rheumatoid arthritis, systemic lupus erythematosus and psoriasis, have an increased risk of arterial and venous thromboembolic events (4). During the past decade, several studies have suggested that patients with inflammatory bowel disease (IBD), including ulcerative colitis (UC) and Crohn's disease (CD), might be at an increased risk for CVDs including coronary artery disease (CAD) and arterial and venous thromboembolic events (5-9).

Endothelial dysfunction is critically involved in the development and progression of atherosclerosis and related vascular diseases. Noninvasive vascular function tests such as flow-mediated dilatation (FMD), nitroglycerin-mediated dilatation (NMD), reactive hyperemia assessed by peripheral arterial tonometry (RH-PAT) have been used to evaluate endothelial function *in vivo* (10). The assessment of endothelial function via brachial artery FMD has been widely used in human subjects. One of the principal mediators for FMD is endothelium-derived nitric oxide (NO) (11). NMD, an index of endothelium-independent vasodilation, assessed by sublingual administration of nitroglycerine, has been used to determine if impairment in vasodilation is due to a loss in smooth-muscle cell integrity, or the inability of endothelial cells to release NO (12). RH-PAT does not directly measure vasodilation, rather, augmentation in finger pulse pressure, which is believed to reflect microvascular dilation through both endothelium-dependent and -independent mechanisms (13).

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Arterial stiffness is a growing epidemic associated with increased risk of CVDs and assessed primarily using pulse wave velocity (PWV) and augmentation index (AIx). Increased arterial stiffness is a well-established indicator for vascular endothelial dysfunction and an independent predictor of cardiovascular events (14). AIx has also been shown to be associated with increased cardiovascular risk (15). One of the important features for arterial atherosclerosis is smooth muscle hyperplasia with increased carotid intima-media thickness (cIMT), a strong predictor of future vascular events including myocardial infarction and stroke (16). Although there are numerous studies evaluating endothelial function, arterial stiffness, and carotid intima-media thickness in IBD patients (17, 18), there is no comprehensive study to systematically review and summarize the data from these individual studies.

Objective: The objective of this review is to systematically evaluate endothelial function, arterial stiffness and carotid intima-media thickness in patients with IBD.

Methods

Patients and public involvement

No patient involved.

Study selection

Observational studies (case-controlled, cross-sectional or cohort) will be included if meeting the following criteria: (1) investigating endothelial function and arterial stiffness or their surrogate markers in patients with IBD (including patients with UC or CD) and controls; (2) reporting measurements of cIMT in patients with IBD and control subjects; (3) written in English. Every effort will be made to include all the studies that meet these criteria.

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Search strategies

A systematic review following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (19) has been prospectively registered with PROSPERO (protocol number CRD42021274093). A comprehensive literature search and review will be conducted through the electronic databases for published studies at PubMed (1945-present), EMBASE (1974-present), Cochrane library (1996-present) and Web of Science (1900-present). We will use the Medical Subject Heading (MeSH) terms and key words in all possible combinations using Boolean operators with the following search strategies: "Inflammatory Bowel Diseases", "Inflammatory Bowel Disease", "Bowel Diseases, Inflammatory", "IBD", "Ulcerative Colitis", "Colitis, Ulcerative", "Idiopathic Proctocolitis", "Colitis Gravis", "Inflammatory Bowel Disease, Ulcerative Colitis Type", "UC", "Crohn Disease", "Crohn's Disease", "Crohn's Enteritis", "CD", "endothelium-dependent flow-mediated dilatation", "endothelium-independent flow mediated dilatation", "FMD", "nitroglycerin-mediated dilatation", "NMD", "endothelial function", "peripheral arterial tonometry", "PAT", "reactive hyperemia", "vascular stiffness", "arterial stiffness", "pulse wave velocity", "PWV", "augmentation index", "AIx", "intima-media thickness" and "IMT" (see **supplementary appendix 1**). The references of all full-text articles will be thoroughly reviewed to identify other potentially relevant articles. All articles will be initially screened and evaluated according to their titles and abstracts. Articles describing animal studies only and non-English language articles will be excluded. Subsequently, the full articles will be evaluated by two independent reviewers (HW and TH) for eligibility. Articles will be downloaded into Endnote and any duplicates will be deleted.

Data extraction

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Two independent investigators (HH and MX) will screen each article and extract the data. In case of disagreement, a third investigator will be consulted (MAH). Discrepancies will be resolved by consensus. Information will be collected on study design, country of origin of the studies, sample size, major clinical and demographic variables, techniques for assessing endothelial function and arterial stiffness and IMT, as well as study outcomes associated with FMD, NMD, RH-PAT, PWV, AIx, and cIMT.

Quality assessment

Based on the characteristics of the included studies, the methodological quality of each study will be evaluated and scored by 2 independent investigators (CX and ZL) using the Newcastle-Ottowa Scale (NOS) (20), which is specifically developed to assess the quality of nonrandomized observational studies. The scoring system consists of 3 major components: 1) the selection of the study groups; 2) the comparability of the groups; 3) and the ascertainment of either the exposure or outcome of interest for case-control or cohort studies respectively. The total scores for each study range between 0 and 9, a higher score representing better methodological quality.

Outcomes

The outcomes of the review will be grouped into the following headings:

Endothelial function: will include direct measurements of FMD, NMD or RH-PAT.

Arterial stiffness: will be assessed by central PWV, peripheral PWV, brachial-ankle PWV and AIx.

Intima-media thickness: will primarily include carotid IMT. If available, aortic or femoral arterial IMT will also be included.

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Analysis

Descriptive analysis

A narrative synthesis of the outcomes of the selected studies will be presented in the final review. This will include the following:

1. Type of study design, control group and sample size;

2. Major clinical and demographic variables; disease activity (active or inactive), disease extent and illness behavior (Montreal classification)(21), if possible;

3. Techniques for assessing endothelial function, arterial stiffness and IMT;

4. Study outcomes: this will include the change in FMD, NMD, RH-PAT, PWV, AIx, and cIMT.

Statistical analysis

Meta-analysis (where possible) will be conducted to assess endothelial function, arterial stiffness and cIMT in patients with IBD. Difference between patients and controls will be calculated using the standardized mean difference (SMD) for continuous variables. These estimates and their 95% confidence intervals (95% CI) will be illustrated using a forest plot. The overall effect sizes will be estimated using both fixed-effects models and random-effects models. Statistical heterogeneity will be calculated using Higgin's (I^2) tests. Subgroup analyses, conditional on the number of studies and their sample sizes, will be stratified based on disease category (eg, UC/CD) or sex (male/female) or disease activity (e.g., active/inactive).

Potential publication and small sample size bias will be assessed by visual inspections of funnel plots and Egger's test. To explore the effect of main factors of interest on predicting SMD, meta-regression analysis will be conducted with restricted maximum likelihood (ReML) estimation method, controlling for potential confounders including disease activity, treatment, and coexistence of traditional cardiovascular risk factors. All statistical analyses will be conducted

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using R package metafor (22). A p value < 0.05 will be considered statistically significant for all analyses.

Conclusion

This systematic review will provide evidence to test the hypothesis that patients with IBD have an elevated CVD burden. This conclusion will stem from direct measurements of endothelial function, arterial stiffness and cIMT in patients with IBD and control subjects. If sufficient data are available, we will conduct a meta-analysis to determine the relationship between patients with IBD and the CVD risk surrogates detailed above. We will also examine if sex, disease activity and treatment have significant impacts on these CVD risk surrogate markers. Overall, the review will provide comprehensive information on endothelial function and CVD in patients with IBD.

Acknowledgments: None.

Author contributions: HW, ZL and HH conceived the idea, planned, and designed the study protocol. HW wrote the first draft; TH, HH and MX planned the data extraction and statistical analysis; CX and ZL provided critical insights. MAH and ZL critically reviewed and modified the manuscript. All authors have reviewed and approved the manuscript.

Funding: None.

Competing interests: None declared.

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Amendments: If this protocol needs to be amended, we will provide the date of each amendment, describe the change and the rationale in this section. Changes will not be incorporated into the protocol.

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Supplementary appendix 1

The following Medical Subject Heading (MeSh) terms and Title/Abstracts search will be implemented in PubMed database:

“Inflammatory Bowel Diseases”, “Inflammatory Bowel Disease”, “Bowel Diseases, Inflammatory”, “IBD”, “Ulcerative Colitis”, “Colitis, Ulcerative”, “Idiopathic Proctocolitis”, “Gravis”, “Inflammatory Bowel Disease, Ulcerative Colitis Type”, “UC”, “Crohn Disease”, “Crohn's Disease”, “Crohn's Enteritis”, “CD”, “endothelium-dependent flow-mediated dilatation”, “endothelium-independent flow mediated dilatation”, “FMD”, “nitroglycerin-mediated dilatation”, “NMD”, “endothelial function”, “peripheral arterial tonometry”, “PAT”, “reactive hyperemia”, “vascular stiffness”, “arterial stiffness”, “pulse wave velocity”, “PWV”, “augmentation index”, “AIx”, “intima-media thickness”, “IMT”.

We applied the following Boolean operators to develop the search: “AND” and “OR” retrieving different MeSH Heading/Subheading Combination

("Inflammatory Bowel Diseases"[MeSH Terms] OR "Inflammatory Bowel Disease"[Title/Abstract] OR "Bowel Diseases, Inflammatory"[Title/Abstract] OR "IBD"[Title/Abstract] OR "Colitis, Ulcerative"[MeSH Terms] OR "Ulcerative Colitis"[Title/Abstract] OR "Idiopathic Proctocolitis"[Title/Abstract] OR "Gravis"[Title/Abstract] OR "Inflammatory Bowel Disease, Ulcerative Colitis Type"[Title/Abstract] OR "UC"[Title/Abstract] OR "Crohn Disease"[MeSH Terms] OR "Crohn's Disease"[Title/Abstract] OR "Crohn's Enteritis"[Title/Abstract] OR "CD"[Title/Abstract]) AND ("endothelium-dependent flow-mediated dilatation"[Title/Abstract] OR "endothelium-independent flow mediated dilatation"[Title/Abstract] OR "FMD"[Title/Abstract] OR "nitroglycerin-mediated dilatation"[Title/Abstract] OR "NMD"[Title/Abstract] OR "endothelial function"[Title/Abstract] OR "peripheral arterial tonometry"[Title/Abstract] OR "PAT"[Title/Abstract] OR "reactive hyperemia"[Title/Abstract] OR "vascular stiffness"[MeSH Terms] OR "arterial stiffness"[Title/Abstract] OR "pulse wave velocity"[Title/Abstract] OR "PWV"[Title/Abstract] OR "augmentation index"[Title/Abstract] OR "AIx"[Title/Abstract] OR "intima-media thickness"[Title/Abstract] OR "IMT"[Title/Abstract])

The following Boolean operators to develop the search in Embase: “AND” and “OR” retrieving different MeSH Heading/Subheading Combination

('inflammatory bowel disease'/exp OR 'ulcerative colitis'/exp OR 'crohn disease'/exp OR 'inflammatory bowel disease' OR 'bowel diseases, inflammatory' OR 'ibd' OR 'colitis, ulcerative' OR 'idiopathic proctocolitis' OR 'gravis' OR 'inflammatory bowel disease, ulcerative colitis type' OR 'uc' OR 'crohn disease' OR 'crohn enteritis' OR 'cd' AND 'arterial stiffness'/exp OR 'endothelium-dependent flow-mediated dilatation' OR 'endothelium-independent flow mediated dilatation' OR 'fmd' OR 'nitroglycerin-mediated dilatation' OR 'nmd' OR 'endothelial

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function' OR 'peripheral arterial tonometry' OR 'pat' OR 'reactive hyperemia' OR 'vascular stiffness' OR 'pulse wave velocity' OR 'pwv' OR 'augmentation index' OR 'aix' OR 'intima-media thickness' OR 'imt')

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	NA

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	9
Amendments			
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Sponsor	#5b	Provide name for the review funder and / or sponsor	9
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	9
Introduction			

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Rationale	#6	Describe the rationale for the review in the context of what is already known	3-4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	5
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	5
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	6

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Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	6
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6-7
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6-7
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	7-8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	7-8
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	7-8
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	7-8

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Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	7-8
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	7-8
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	6

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Secondary Subject Heading:	Cardiovascular medicine
Keywords:	Inflammatory bowel disease < GASTROENTEROLOGY, Vascular medicine < INTERNAL MEDICINE, CARDIOLOGY

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Abstract

Introduction: Patients with inflammatory bowel disease (IBD) might be at an increased risk for the development of cardiovascular disease (CVD). The present protocol is developed to review and analyze published data to determine if patients with IBD have an increased CVD burden.

Methods and analysis: We will conduct a systematic review of all observational studies that examine endothelial function, arterial stiffness, and carotid intima-media thickness in IBD patients. Study selection will follow the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, and study quality will be assessed using the Newcastle-Ottawa Scale (NOS). If sufficient data are available, a meta-analysis will be conducted. The overall effect sizes will be estimated using both fixed-effects models and random-effects models. Statistical heterogeneity will be calculated using Higgin's (I^2) tests. Subgroup analyses, conditional number of studies retrieved and their sample size, will be stratified according to participant disease category or gender or disease activity.

Strengths and limitations of this study

- This study employs a broad search strategy from multiple large databases, including PubMed, EMBASE, Cochrane library and Web of Science.
- The data reporting will adhere to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.
- The study may be impacted by significant heterogeneity in the included studies due to differences in diagnostic tests, disease activities, and the presence of traditional cardiovascular risk factors.

Ethics and dissemination: Formal ethics approval is not required as individual data will not be collected. The results will be disseminated through peer-reviewed publications, conference presentations and scientific news releases.

Trial registration number: International Prospective Register for Systematic Reviews (PROSPERO) number CRD42021274093.

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Introduction

Rationale: Cardiovascular disease (CVD) remains the leading cause of morbidity and mortality despite aggressive treatment of traditional risk factors (1). Systemic inflammation and endothelial cell dysfunction are considered among the key factors that are critically involved in the development and progression of CVD (2, 3). Indeed, patients with chronic inflammatory diseases such as rheumatoid arthritis, systemic lupus erythematosus and psoriasis, have an increased risk of arterial and venous thromboembolic events (4). During the past decade, several studies have suggested that patients with inflammatory bowel disease (IBD), including ulcerative colitis (UC) and Crohn's disease (CD), might be at an increased risk for CVDs including coronary artery disease (CAD) and arterial and venous thromboembolic events (5-9).

Endothelial dysfunction is critically involved in the development and progression of atherosclerosis and related vascular diseases. Noninvasive vascular function tests such as flow-mediated dilatation (FMD), nitroglycerin-mediated dilatation (NMD), reactive hyperemia assessed by peripheral arterial tonometry (RH-PAT) have been used to evaluate endothelial function *in vivo* (10). The assessment of endothelial function via brachial artery FMD has been widely used in human subjects. One of the principal mediators for FMD is endothelium-derived nitric oxide (NO) (11). NMD, an index of endothelium-independent vasodilation, assessed by sublingual administration of nitroglycerine, has been used to determine if impairment in vasodilation is due to a loss in smooth-muscle cell integrity, or the inability of endothelial cells to release NO (12). RH-PAT does not directly measure vasodilation, rather, augmentation in finger pulse pressure, which is believed to reflect microvascular dilation through both endothelium-dependent and -independent mechanisms (13).

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Arterial stiffness is a growing epidemic associated with increased risk of CVDs and assessed primarily using pulse wave velocity (PWV) and augmentation index (AIx). Increased arterial stiffness is a well-established indicator for vascular endothelial dysfunction and an independent predictor of cardiovascular events (14). AIx has also been shown to be associated with increased cardiovascular risk (15). One of the important features for arterial atherosclerosis is smooth muscle hyperplasia with increased carotid intima-media thickness (cIMT), a strong predictor of future vascular events including myocardial infarction and stroke (16). Although there are numerous studies evaluating endothelial function, arterial stiffness, and carotid intima-media thickness in IBD patients (17, 18), there is no comprehensive study to systematically review and summarize the data from these individual studies.

Objective: The objective of this review is to systematically evaluate endothelial function, arterial stiffness and carotid intima-media thickness in patients with IBD.

Methods

Patients and public involvement

No patient involved.

Study selection

Observational studies (case-controlled, cross-sectional or cohort) will be included if meeting the following criteria: (1) investigating endothelial function and arterial stiffness or their surrogate markers in patients with IBD (including patients with UC or CD) and controls; (2) reporting measurements of cIMT in patients with IBD and control subjects; (3) written in English. Every effort will be made to include all the studies that meet these criteria.

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Search strategies

A systematic review following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (19) has been prospectively registered with PROSPERO (protocol number CRD42021274093). A comprehensive literature search and review will be conducted through the electronic databases for published studies at PubMed (1945-present), EMBASE (1974-present), Cochrane library (1996-present) and Web of Science (1900-present). We will use the Medical Subject Heading (MeSH) terms and key words in all possible combinations using Boolean operators with the following search strategies: "Inflammatory Bowel Diseases", "Inflammatory Bowel Disease", "Bowel Diseases, Inflammatory", "IBD", "Ulcerative Colitis", "Colitis, Ulcerative", "Idiopathic Proctocolitis", "Colitis Gravis", "Inflammatory Bowel Disease, Ulcerative Colitis Type", "UC", "Crohn Disease", "Crohn's Disease", "Crohn's Enteritis", "CD", "endothelium-dependent flow-mediated dilatation", "endothelium-independent flow mediated dilatation", "FMD", "nitroglycerin-mediated dilatation", "NMD", "endothelial function", "peripheral arterial tonometry", "PAT", "reactive hyperemia", "vascular stiffness", "arterial stiffness", "pulse wave velocity", "PWV", "augmentation index", "AIx", "intima-media thickness" and "IMT" (see **supplementary appendix 1**). The references of all full-text articles will be thoroughly reviewed to identify other potentially relevant articles. All articles will be initially screened and evaluated according to their titles and abstracts. Articles describing animal studies only and non-English language articles will be excluded. Subsequently, the full articles will be evaluated by two independent reviewers (HW and TH) for eligibility. Articles will be downloaded into Endnote and any duplicates will be deleted. Data collection is anticipated to start in Feburary of 2022 and to be completed by October 2022.

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Data extraction

Two independent investigators (HH and MX) will screen each article and extract the data. In case of disagreement, a third investigator will be consulted (MAH). Discrepancies will be resolved by consensus. Information will be collected on study design, country of origin of the studies, sample size, major clinical and demographic variables, techniques for assessing endothelial function and arterial stiffness and IMT, as well as study outcomes associated with FMD, NMD, RH-PAT, PWV, AIx, and cIMT.

Quality assessment

Based on the characteristics of the included studies, the methodological quality of each study will be evaluated and scored by 2 independent investigators (CX and ZL) using the Newcastle-Ottowa Scale (NOS) (20), which is specifically developed to assess the quality of nonrandomized observational studies. The scoring system consists of 3 major components: 1) the selection of the study groups; 2) the comparability of the groups; 3) and the ascertainment of either the exposure or outcome of interest for case-control or cohort studies respectively. The total scores for each study range between 0 and 9, a higher score representing better methodological quality.

Outcomes

The outcomes of the review will be grouped into the following headings:

Endothelial function: will include direct measurements of FMD, NMD or RH-PAT.

Arterial stiffness: will be assessed by central PWV, peripheral PWV, brachial-ankle PWV and AIx.

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Intima-media thickness: will primarily include carotid IMT. If available, aortic or femoral arterial IMT will also be included.

Analysis

Descriptive analysis

A narrative synthesis of the outcomes of the selected studies will be presented in the final review. This will include the following:

1. Type of study design, control group and sample size;

2. Major clinical and demographic variables; disease activity (active or inactive), disease extent and illness behavior (Montreal classification)(21), if possible;

3. Techniques for assessing endothelial function, arterial stiffness and IMT;

4. Study outcomes: this will include the change in FMD, NMD, RH-PAT, PWV, AIx, and cIMT.

Statistical analysis

Meta-analysis (where possible) will be conducted to assess endothelial function, arterial stiffness and cIMT in patients with IBD. Difference between patients and controls will be calculated using the standardized mean difference (SMD) for continuous variables. These estimates and their 95% confidence intervals (95% CI) will be illustrated using a forest plot. The overall effect sizes will be estimated using both fixed-effects models and random-effects models. Statistical heterogeneity will be calculated using Higgin's (I^2) tests. Subgroup analyses, conditional on the number of studies and their sample sizes, will be stratified based on disease category (eg, UC/CD) or sex (male/female) or disease activity (e.g., active/inactive).

Potential publication and small sample size bias will be assessed by visual inspections of funnel plots and Egger's test. To explore the effect of main factors of interest on predicting SMD,

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meta-regression analysis will be conducted with restricted maximum likelihood (ReML) estimation method, controlling for potential confounders including disease activity, treatment, and coexistence of traditional cardiovascular risk factors. All statistical analyses will be conducted using R package metafor (22). A p value < 0.05 will be considered statistically significant for all analyses.

Acknowledgments: None.

Author contributions: HW, ZL and HH conceived the idea, planned, and designed the study protocol. HW wrote the first draft; TH, HH and MX planned the data extraction and statistical analysis; CX and ZL provided critical insights. MAH and ZL critically reviewed and modified the manuscript. All authors have reviewed and approved the manuscript.

Funding: None.

Competing interests: None declared.

Amendments: If this protocol needs to be amended, we will provide the date of each amendment, describe the change and the rationale in this section. Changes will not be incorporated into the protocol.

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Supplementary appendix 1

The following Medical Subject Heading (MeSh) terms and Title/Abstracts search will be implemented in PubMed database:

“Inflammatory Bowel Diseases”, “Inflammatory Bowel Disease”, “Bowel Diseases, Inflammatory”, “IBD”, “Ulcerative Colitis”, “Colitis, Ulcerative”, “Idiopathic Proctocolitis”, “Gravis”, “Inflammatory Bowel Disease, Ulcerative Colitis Type”, “UC”, “Crohn Disease”, “Crohn's Disease”, “Crohn's Enteritis”, “CD”, “endothelium-dependent flow-mediated dilatation”, “endothelium-independent flow mediated dilatation”, “FMD”, “nitroglycerin-mediated dilatation”, “NMD”, “endothelial function”, “peripheral arterial tonometry”, “PAT”, “reactive hyperemia”, “vascular stiffness”, “arterial stiffness”, “pulse wave velocity”, “PWV”, “augmentation index”, “AIx”, “intima-media thickness”, “IMT”.

We applied the following Boolean operators to develop the search: “AND” and “OR” retrieving different MeSH Heading/Subheading Combination

("Inflammatory Bowel Diseases"[MeSH Terms] OR "Inflammatory Bowel Disease"[Title/Abstract] OR "Bowel Diseases, Inflammatory"[Title/Abstract] OR "IBD"[Title/Abstract] OR "Colitis, Ulcerative"[MeSH Terms] OR "Ulcerative Colitis"[Title/Abstract] OR "Idiopathic Proctocolitis"[Title/Abstract] OR "Gravis"[Title/Abstract] OR "Inflammatory Bowel Disease, Ulcerative Colitis Type"[Title/Abstract] OR "UC"[Title/Abstract] OR "Crohn Disease"[MeSH Terms] OR "Crohn's Disease"[Title/Abstract] OR "Crohn's Enteritis"[Title/Abstract] OR "CD"[Title/Abstract]) AND ("endothelium-dependent flow-mediated dilatation"[Title/Abstract] OR "endothelium-independent flow mediated dilatation"[Title/Abstract] OR "FMD"[Title/Abstract] OR "nitroglycerin-mediated dilatation"[Title/Abstract] OR "NMD"[Title/Abstract] OR "endothelial function"[Title/Abstract] OR "peripheral arterial tonometry"[Title/Abstract] OR "PAT"[Title/Abstract] OR "reactive hyperemia"[Title/Abstract] OR "vascular stiffness"[MeSH Terms] OR "arterial stiffness"[Title/Abstract] OR "pulse wave velocity"[Title/Abstract] OR "PWV"[Title/Abstract] OR "augmentation index"[Title/Abstract] OR "AIx"[Title/Abstract] OR "intima-media thickness"[Title/Abstract] OR "IMT"[Title/Abstract])

The following Boolean operators to develop the search in Embase: “AND” and “OR” retrieving different MeSH Heading/Subheading Combination

('inflammatory bowel disease'/exp OR 'ulcerative colitis'/exp OR 'crohn disease'/exp OR 'inflammatory bowel disease' OR 'bowel diseases, inflammatory' OR 'ibd' OR 'colitis, ulcerative' OR 'idiopathic proctocolitis' OR 'gravis' OR 'inflammatory bowel disease, ulcerative colitis type' OR 'uc' OR 'crohn disease' OR 'crohn enteritis' OR 'cd' AND 'arterial stiffness'/exp OR 'endothelium-dependent flow-mediated dilatation' OR 'endothelium-independent flow mediated dilatation' OR 'fmd' OR 'nitroglycerin-mediated dilatation' OR 'nmd' OR 'endothelial

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function' OR 'peripheral arterial tonometry' OR 'pat' OR 'reactive hyperemia' OR 'vascular stiffness' OR 'pulse wave velocity' OR 'pwv' OR 'augmentation index' OR 'aix' OR 'intima-media thickness' OR 'imt')

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	NA

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	9
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	9
Support			
Sources	#5a	Indicate sources of financial or other support for the review	9
Sponsor	#5b	Provide name for the review funder and / or sponsor	9
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	9
Introduction			

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Rationale	#6	Describe the rationale for the review in the context of what is already known	3-4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	5
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	5
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	6

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Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	6
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6-7
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6-7
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	7-8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	7-8
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	7-8
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	7-8

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Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	7-8
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	7-8
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	6

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